

COMP4131: Data Modelling and Analysis

Lecture 4: Analysis and Modelling

Kian Ming Lim

University of Nottingham Ningbo China

kian-ming.lim@nottingham.edu.cn

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Outline

- 1 Data Analysis
- 2 EDA before Modelling
- 3 Data Modelling
- 4 From Problem to Modelling Task
- 5 Modelling Workflow

Recap of Lectures 1 to 3

What we have already covered

- Lecture 1: DMA workflow and module overview
- Lecture 2: data cleaning, transformation, encoding, scaling
- Lecture 3: visualisation for exploration and communication

What is still missing?

- Turning a vague question into a modelling task 将模糊问题转换为建模问题
- Deciding what to predict and how to evaluate it
- Building a defensible analysis-to-modelling pipeline

Why this lecture matters

- Many failed data science projects do not fail because of a weak algorithm.
- They fail because the **problem is unclear**, the **data is misunderstood**, or the **evaluation is inappropriate**.
- Technical implementation is important, but correct **problem framing** is equally important.

Key takeaway

Modelling is not just about running an algorithm. It is about making a sequence of sensible decisions.

By the end of this lecture, you should be able to:

- ① distinguish between data analysis and data modelling;
- ② translate a real-world question into a suitable modelling task;
- ③ perform pre-modelling analytical checks using EDA;
- ④ describe a robust modelling workflow;
- ⑤ select suitable evaluation metrics for different task types;
- ⑥ identify common pitfalls such as leakage, overfitting, and poor target definition.

Data Analysis

What is data analysis?

Definition

Data analysis is the process of ^{检查}inspecting, summarising, exploring, and ^{解释}interpreting data to discover structure, ^{模式}patterns, ^{异常值}anomalies, and relationships.

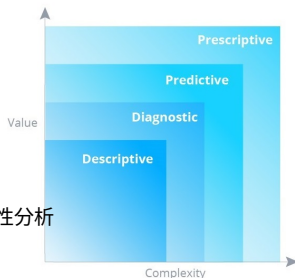
- It helps us understand **what is in the data.**
- It helps us decide **what to do next.**
- It often leads to hypotheses, design choices, or modelling decisions.
_{假设} _{建模决策}

Overview of Data Analysis Techniques

Purpose: To transform raw data into actionable insights. 可执行洞察

Types of Data Analysis:

- Descriptive Analysis
- Diagnostic Analysis 诊断性分析
- Predictive Analysis
- Prescriptive Analysis 规范性分析
- Exploratory Data Analysis (EDA) 探索性数据分析



Descriptive analytics addresses the issue of what happened.

Diagnostic analytics answers the question of why something happened.

Predictive analytics describes what is likely to happen.

Prescriptive analytics prescribes what step to take to avoid a future problem.

Source: Things You Should Know About Types Of Data Analysis

Descriptive Analysis

Definition:

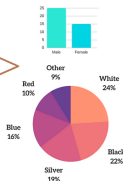
- Goal: summarise the main characteristics of a dataset. 概括数据集的主要特征
- Typical outputs: means, medians, distributions, frequencies, dashboards. 典型输出：均值、中位数、分布、频率、仪表盘
- Often the **first step** in every analytical workflow.

Examples:

- What is the average coursework mark?
- Which product category has the highest sales volume?
- How many students attended at least 80% of classes?

	A	B	C	D
1	Respondent Number	Age	Gender	Favorite Car Color
2	1	22	M	White
3	2	37	F	Silver
4	3	45	F	Black
5	4	62	F	Gray
6	5	28	M	Red
7	6	45	M	Green
8	7	88	F	Brown
9	8	61	M	White
10	9	99	M	Black
11	10	27	M	White
12	11	39	F	Green
13	12	43	M	Brown
14	13	55	F	Black
15	14	58	F	White

RAW DATA



Descriptive Statistics

Source: Descriptive Statistics Examples, Types and Definition

Diagnostic Analysis

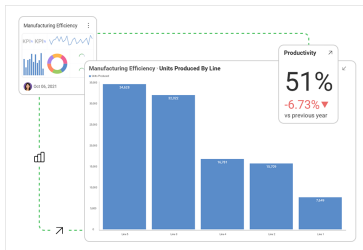
Definition:

探究可能解释观察到的结果的因素--》解释为什么发生

- Goal: investigate factors that may explain an observed outcome.
- Typical tools: segmentation, comparison, drill-down analysis, correlation, domain reasoning. 常用工具：细分、比较、深入分析、相关性分析、领域推理

Examples:

- **Why** did server response time increase last week?
- Why are some customers abandoning their baskets?
- Why did one class perform worse than another?



Source: Types of Data Analysis, Benefits & Examples

Predictive Analysis

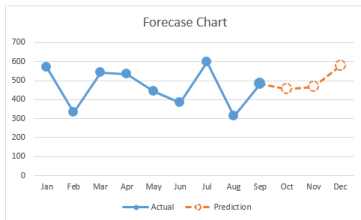
Definition:

根据观测数据预测未知或未来的结果

- Goal: estimate unknown or future outcomes from observed data.
- This is where many machine learning models are used.
- Prediction is useful even when full causal understanding is unavailable.

Examples:

- Sales forecasting
- Customer ^{流失} churn prediction
- Predicting student pass/fail
- Forecasting CPU load over the next hour



Source: Forecast Chart.

Definition:

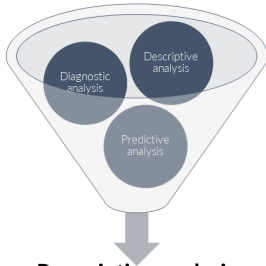
基于数据洞察提出行动建议，而不仅仅是描述或预测结果--》建议行动

- Goal: recommend actions based on data insights rather than merely describe or predict outcomes.
- Often combines predictions with **constraints, costs, or optimisation criteria**. 通常将预测与约束条件、成本或优化标准相结合
- It is common in **operations research, decision support, and automated systems**. 常见于运筹学、决策支持和自动化系统中

Examples:

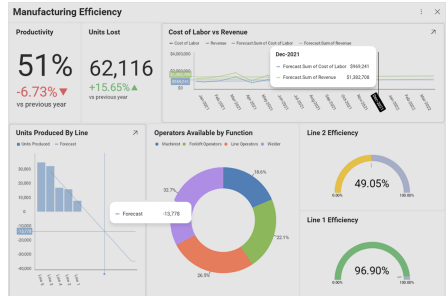
- Which intervention should a university provide to at-risk students?
- Which ad campaign should a company choose given a fixed budget?
- How should cloud resources be allocated to meet demand at minimal cost?

Prescriptive Analysis



Prescriptive analysis www.lido.app

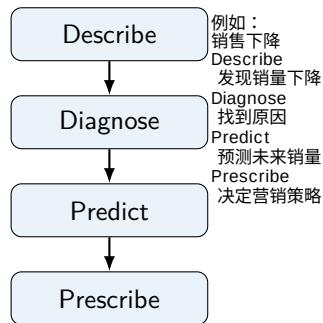
Source: Data Analysis 101



Source: Types of Data Analysis, Benefits & Examples

Analysis in practice

- Real projects **rarely use only one type** of analysis. 实际项目很少只使用一种分析方法。
- Descriptive analysis **often reveals patterns**. 描述性分析通常揭示模式
- Diagnostic analysis **explains those patterns**. 诊断性分析解释这些模式。
- Predictive analysis **formalises them**. 预测性分析将这些模式形式化
- Prescriptive analysis **turns them into action**. 规范性分析将这些模式转化为行动



EDA before Modelling

Why exploratory data analysis matters

EDA helps us to:

- understand the structure of the dataset;
- identify quality issues early;
- find relationships and potential signal;
- guide feature engineering and model selection;
- detect risks before formal modelling begins.

Before modelling, check:

- dataset size
- variable types 变量类型
- missing values
- duplicates 重复值
- outliers 异常值
- class balance 类别平衡
- target distribution 目标分布
- pairwise relationships 成对关系

Dataset Shape and Structure

Questions to ask:

- How many **rows and columns** are there?
- Is the sample size **adequate** for the task?
- Is the dataset **wide or tall**?
- Are there too many **sparse features**?

Interpretation

A dataset with very few rows but many columns may **be harder to model reliably**, especially when there are **many sparse or weak features**.

行数很少但列数很多的数据集可能难以可靠地建模，尤其是在存在许多稀疏或弱特征的情况下

Example: Dataset Shape and Structure

Dataset	Rows	Columns
Student Marks	500	8
Customer Survey	120	250
Log Events	50,000	12



Tall



Wide

Tall dataset: many rows, fewer columns

Wide dataset: fewer rows, many columns

Data Types and Formatting

- Are numbers stored as **text**?
- Are dates **correctly parsed**?
- Are categories **consistently named**?
- Are ID columns **being mistakenly used as predictors**?

Example

ZIP code may look numeric, but it should usually be treated **as a categorical location code**.

Example: Data Types and Formatting

ID	Age	Date	City
1001	"23"	01/02/26	NB
1002	"NA"	2026-02-01	Ningbo
1003	"35"	Feb 3, 2026	ningbo

- Age is stored as text rather than numeric.
- Dates use inconsistent formats.
- City labels are inconsistent.
- ID should usually not be used as a predictor.

Missing Values

- Missingness can be random or systematic.
- Systematic missingness may itself carry information.
- Common choices include deletion, imputation, or explicit missing-value indicators.
插补 显示缺失值指示

Example: Missing Values

Student	Attendance	Quiz	Final Mark
A	92	71	76
B	–	55	57
C	88	–	84
D	45	40	–
E	76	63	69



observed
missing

Missing final marks may indicate absence or withdrawal, not merely random loss of data.

最终成绩缺失可能表示缺席或退学，而不仅仅是数据随机丢失。

Outliers and Unusual Observations

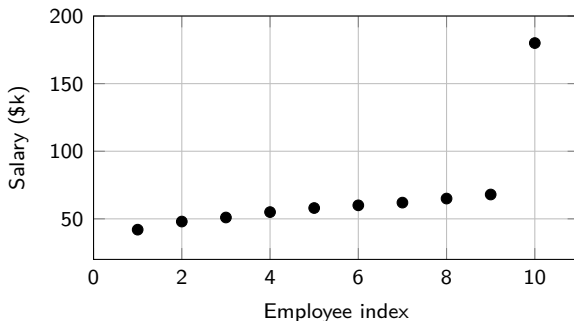
- Some outliers are data entry errors. 数据录入错误
- Some outliers are valid but rare cases. 有效但罕见的案例
- Some outliers are precisely the cases we care most about. 恰恰是我们最关注的案例

Key takeaway

Do not remove outliers automatically. First ask why they exist.

不要自动删除异常值。首先要探究它们存在的原因

Example: Outliers and Unusual Observations



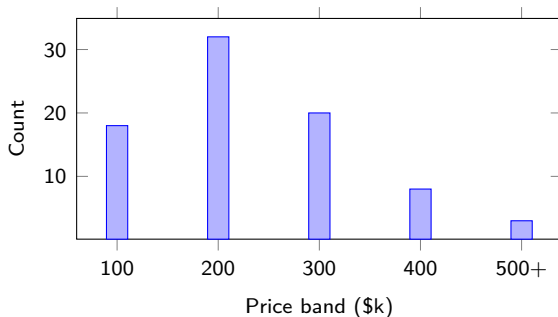
Most salaries are between \$40k and \$70k, but one observation at \$180k is unusual.

- possible data entry error
- senior executive salary
- important rare case

Target Distribution

- For regression, inspect range, skewness, and extreme values.
对于回归分析，请检查极差、偏度和极值
- For classification, inspect class balance and rarity of positive cases.
对于分类分析，请检查类别平衡性和正例稀有度。
- Target distribution strongly affects metric choice and model behaviour.
目标分布对指标选择和模型行为有显著影响。

Example: Regression Target Distribution



右偏分布

This distribution is **right-skewed**, with most houses concentrated in the lower price bands and only a few very expensive properties.

Example: Classification Target Distribution

Class	Count
Non-fraud	950
Fraud	50

This is a highly imbalanced target distribution.

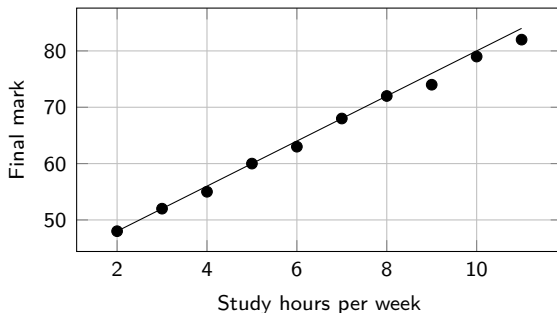
Important implication

A model with 95% accuracy may still be useless if it predicts every case as non-fraud.

Relationship Analysis

- Correlation matrices 相关矩阵
- Scatter plots 散点图
- Box plots by class 按类别绘制箱线图
- Grouped statistics 分组统计
- Pairwise or multivariate relationships 成对/多变量关系

Example: Relationship Analysis



This suggests a positive relationship, but we should still examine attendance, prior knowledge, and assessment difficulty.

Correlation is not Causation

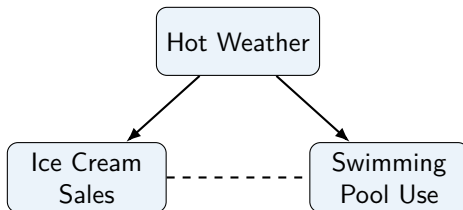
相关不是因果关系

高相关性仅表明存在关联，而非因果关系

- A high correlation indicates association, not proof of cause.
- Hidden variables may explain the pattern. 隐藏变量可能解释这种模式
- Reverse causality is possible. 反向因果关系是可能的
- Some causal effects are non-linear and may not appear strongly correlated.

某些因果效应是非线性的，可能看起来并不具有很强的相关性

Example: Correlation is not Causation



Ice cream sales and swimming pool use may be correlated, but one does not cause the other. Both are influenced by hot weather.

Multicollinearity

多重共线性

两个或多个预测变量可能包含高度重叠的信息

- Two or more predictors may carry highly overlapping information.
- This can affect interpretability and stability, especially for linear models. 这会影响模型的可解释性和稳定性，尤其对于线性模型而言。
- It may not always damage predictive performance, but it complicates explanation.

多重共线性并非总是会损害预测性能，但会使解释变得复杂。

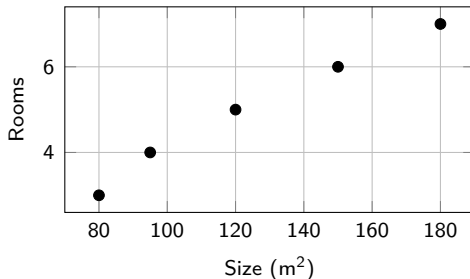
Example

House size in square metres and number of rooms may both track the same underlying factor: property scale.

房屋面积（平方米）和房间数量可能都反映了同一个潜在因素：房产规模。

Example: Multicollinearity

House	Size (m ²)	Rooms
A	80	3
B	95	4
C	120	5
D	150	6
E	180	7



These predictors are strongly related, so their individual coefficients may become unstable in a linear regression model.

Data Leakage during Analysis

数据泄漏

- Using future information when predicting the past
- Including a variable that directly contains the answer
- ^{插补}Imputing or ^{缩放}scaling using the full dataset before ^{拆分}splitting
- Feature selection performed on **n all data before evaluation**

Why it is dangerous

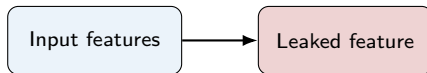
Leakage makes a model appear **unrealistically strong**. It often **fails badly on real unseen data**.

数据泄露会使模型看起来异常强大。它通常会在真实未见过的数据上表现糟糕

Example: Data Leakage during Analysis

Student	Coursework	Final Exam	Total Mark
A	74	78	76
B	58	56	57
C	82	86	84
D	49	43	46

If the task is to predict **Final Exam**, then using **Total Mark** as an input leaks information **because Total Mark already includes the Final Exam score**.



Data Modelling

What is data modelling?

Definition

A model is a mathematical or computational representation that captures useful structure in data for explanation, estimation, classification, prediction, or decision support.

模型是一种数学或计算表示，它捕捉数据中有用的结构，用于解释、估计、分类、预测或决策支持

- Models simplify reality.
- Models make assumptions. 假设
- Models are useful when they generalise to unseen cases.

Why do we build models?

- To predict outcomes for unseen cases
- To estimate relationships between variables
- To automate decision support
- To compare alternative hypotheses or representations
比较不同的假设或表述
- To support planning and optimisation

A model is a simplified view of reality

- No model captures every real-world factor.
- Simpler models may be more interpretable and robust. 更易于解释且更稳健
- Complex models may fit data better but can be harder to explain.
复杂的模型可能更符合数据，但更难解释
- The best model depends on the objective, data, and constraints 约束条件

Famous idea

“All models are wrong, but some are useful.” — George Box

From Problem to Modelling Task

Real-world problems are usually vague

模糊

Examples of vague problem statements:

- Which students are at risk?
- Which customers may leave?
- What affects household energy usage?
- Can we find meaningful behaviour groups?

These are useful starting points, but they are not yet precise modelling tasks. 这些都是有用的起点，但它们还不是精确的建模任务

Why vague questions are a problem

They usually **do not specify**:

- the target variable;
- the input features;
- the prediction horizon; 预测范围
- the unit of analysis; 分析单位
- the evaluation metric;
- the operational meaning of success. 成功的操作性定义

Translating a problem into a modelling task

A good modelling formulation should answer:

- ① What exactly do we want to predict or analyse?
- ② For which entities or observations?
- ③ From which variables?
- ④ Within what time frame?
- ⑤ How will success be measured?

Common modelling task types

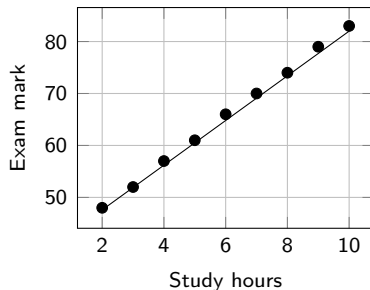
- Regression 回归
- Classification 分类
- Clustering 聚类
- Anomaly detection 异常检测
- Dimensionality reduction 降维
- Time series forecasting 时序预测

Regression

- Use regression when **the target is numerical.** 数值
- The aim is to **predict a quantity on a continuous scale.** 预测连续变量

Examples

- house price prediction
- salary estimation 薪资估计
- exam mark prediction
- energy demand estimation



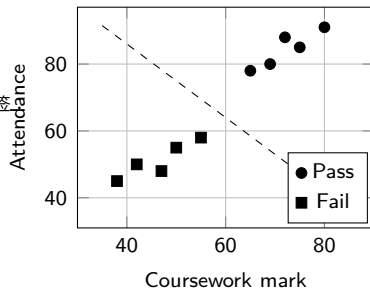
A regression model predicts a continuous value, such as an exam mark.

Classification

- Use classification when the target is **categorical**.
- The aim is to **assign one of a finite set of labels**. 为以下有限个标签分配一个标签

Examples

- pass or fail
- spam or not spam 垃圾邮件还是非垃圾邮件
- churn or no churn 用户流失还是不流失
- benign or malignant 良性还是恶性



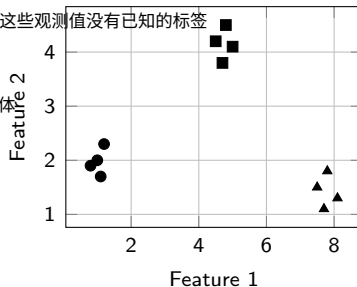
A classification model separates observations into categories such as pass and fail.

Clustering

- Clustering groups observations without known labels. 聚类分析将观测值分组，这些观测值没有已知的标签
- It is useful for discovering hidden structure or segments. 发现隐藏的结构或细分群体

Examples

- customer segmentation
- user behaviour profiling
- document grouping



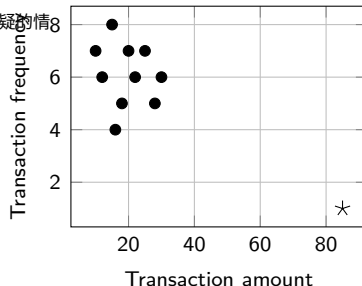
Clustering reveals natural groups in the data without pre-defined labels.

Anomaly Detection

- The goal is to identify unusual, rare, or suspicious cases. 识别异常、罕见或可疑的情况
- In many applications, the abnormal cases are the most important ones.

Examples

- fraud detection 欺诈检测
- network intrusion detection 网络入侵检测
- unusual sensor failure patterns 异常传感器故障模式



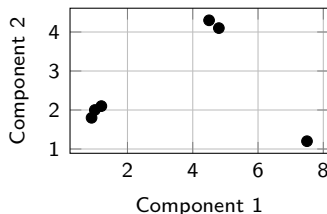
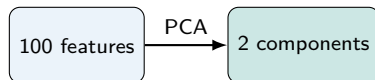
Most points follow a normal pattern; the isolated point may be an anomaly.

Dimensionality Reduction

- Reduce many variables into fewer informative components.
- Useful for visualisation, compression, denoising, and pre-processing. 可视化、压缩、去噪和预处理

Examples

- PCA for dimensionality reduction
- t-SNE or UMAP for visualising high-dimensional embeddings

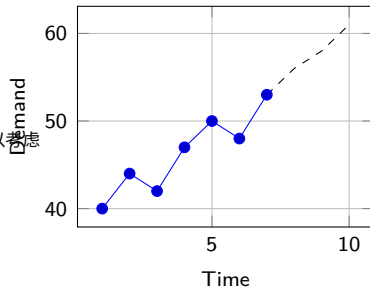


Time Series Forecasting

- Forecasting predicts future values from ordered temporal data. 预测利用有序的时间序列数据来预测未来值
- Time dependency and ordering must be respected during evaluation. 时间依赖性和顺序性在评估过程中必须加以考虑

Examples

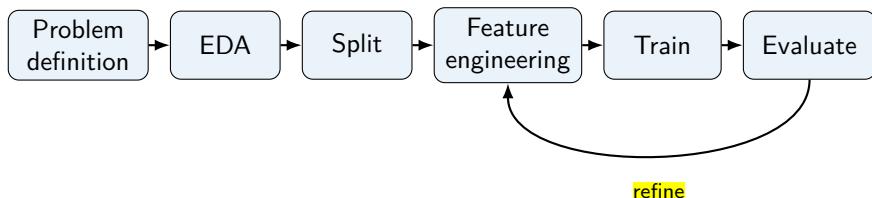
- server load forecasting 服务器负载预测
- electricity demand prediction 电力需求预测
- stock or sales forecasting 库存或销售预测



The solid line shows observed values; the dashed line shows future forecasts.

Modelling Workflow

Basic Modelling Workflow



Key takeaway

Good modelling is iterative, and preprocessing should be learned from the training data only.

Step 1: Problem definition

A good problem definition should specify:

- ① objective;
- ② target variable;
- ③ data source;
- ④ time frame;
- ⑤ task type;
- ⑥ constraints and costs of mistakes.

约束条件

Step 2: Data splitting



- Training set: used to fit the model and learn preprocessing steps.
- Validation set: used to tune hyperparameters and compare model variants.
- Test set: kept unseen until the end for final evaluation.

Why split before feature engineering?

- The validation and test sets should represent truly unseen data.
- If preprocessing is done on the full dataset first, information may leak into training. 泄漏
- This can make performance look better than it really is.

Rule

Split first, then fit preprocessing and feature engineering using the training set only.

Step 3: Feature selection and engineering

- Choose variables likely to carry useful signal.
- Transform raw variables into more informative representations.
- Encode categories, scale numerical features where needed, and create domain-informed features.
- Fit these transformations on the training set, then apply them to validation and test sets.

Examples

- standardising age or income using training-set statistics
- one-hot encoding categorical variables
- creating total weekly study time from daily logs
- extracting text length from free-text responses

Why train and test must be separated

- Training performance **mainly measures fit to seen data.**
- The real goal is performance **on unseen data.**
- **Testing on the same data used for training produces over-optimistic results.**

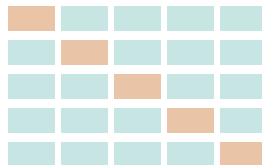
Rule

Never report final performance using data that influenced model fitting, preprocessing, or model selection.

切勿使用影响模型拟合、预处理或模型选择的数据来报告最终性能

Cross-validation

- Useful when the dataset is small.
- Repeatedly splits the training data into training and validation folds.
- Produces a more stable estimate than a single validation split.
- Feature engineering should be fitted separately inside each fold.

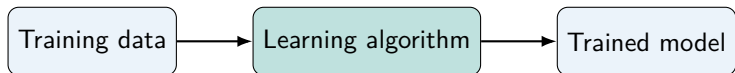


Each row shows one round: one fold is used for validation and the remaining folds are used for training.

每一行显示一轮：一集用于验证，其余集用于训练

Step 4: Train a model

- Use the training set to fit model parameters.
- The algorithm learns relationships between features and target values.
- The trained model can then be applied to validation or test data.



Key takeaway

Training creates a fitted model, but evaluation on unseen data is still essential.

The Formal Learning Problem

According to Tom Mitchell (1997):

Definition

A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .

如果一个计算机程序在任务 T 中的性能（以性能指标 P 衡量）随着经验 E 的积累而提高，则称该程序从经验 E 中学习，以改进其在任务类别 T 和性能指标 P 上的学习

- **Task T :** Classification, Regression, Clustering.
- **Performance P :** Accuracy, F1-score, MSE.
- **Experience E :** The training dataset $\mathcal{D}$.

The Abstract Model

In computer science, we define a model as a function:

$$Y = f(X) + \epsilon$$

- Y : Target variable (Output). 目标变量 (输出)
- X : Feature matrix (Inputs). 特征矩阵 (输入)
- f : The function the model tries to learn. 模型试图学习的函数
- ϵ : Irreducible error (noise). 不可约误差 (噪声)

Goal: Minimize the difference between $f(X)$ and the actual Y .
最小化 $f(X)$ 与实际 Y 之间的差异

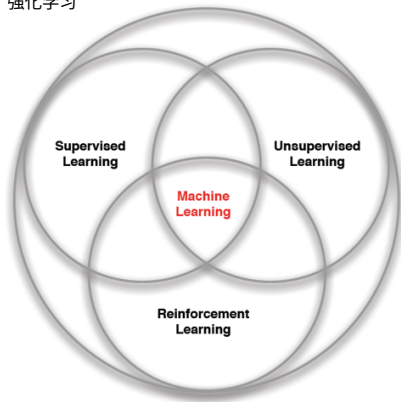
Machine Learning Models

Definition:

- Models that learn patterns from data. 从数据中学习模式的模型

Types:

- Supervised learning, unsupervised learning, reinforcement learning.
监督学习、无监督学习、强化学习

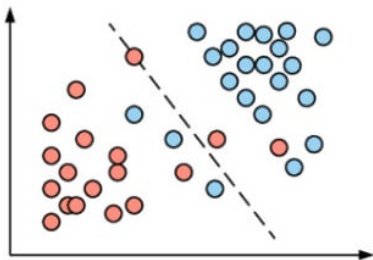


Source: Link

Supervised Learning

Definition: A machine learning approach where the model is trained using labeled data.

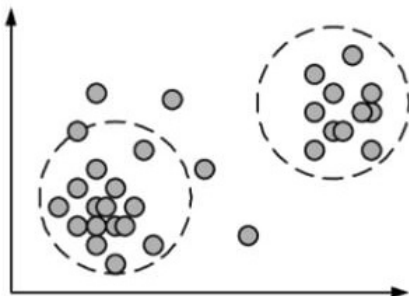
- Input (features) and output (labels) are known.
- Data: (x, y) x is data, y is label
- Learn a function to map $x \rightarrow y$
- Classification, regression, object detection, semantic segmentation, image captioning, etc.
- Examples: Linear regression, logistic regression, decision trees, support vector machines, and many more.



Unsupervised Learning

Definition: A machine learning approach where the model learns patterns **without labeled data.**

- Identifies **hidden structures and relationships in data.**
- **x Just data, no labels!**
- Examples: **K-means clustering, hierarchical clustering, principal component analysis (PCA), and many more.**



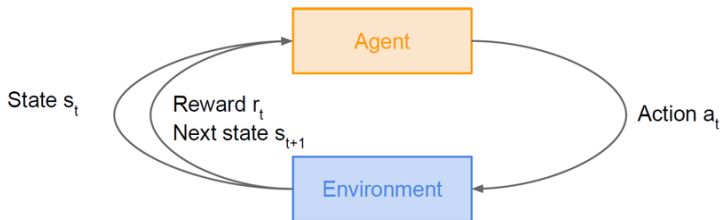
Source: SuperAnnotate

Reinforcement Learning

一种学习范式，其中智能体通过与环境交互并获得奖励或惩罚来学习。

Definition: A learning paradigm where **an agent** learns by **interacting with an environment** and **receiving rewards or penalties**.

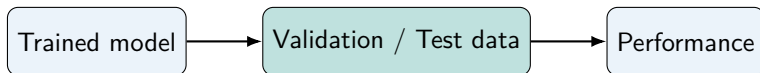
- Used in **robotics, gaming, and autonomous systems**.
- Learn how to take actions in order to **maximize reward**.
- Key components: **Agent, Environment, Reward, Policy**.



Source: Reinforcement Learning

Step 5: Evaluate the model

- After training, evaluate the model on validation or test data.
- The goal is to measure how well the model generalises to unseen samples.
- Choose evaluation metrics that match the task and the real-world objective.



Key takeaway

A good model is not the one that performs best on the training set, but the one that performs well on unseen data.

好的模型并非在训练集上表现最佳的模型，而是在未见过的数据上表现良好的模型。

Evaluation metrics: choose according to the task

Task	Typical metrics
Regression	MAE, MSE, RMSE, R^2
Classification	Accuracy, precision, recall, F1-score, ROC-AUC
Clustering	Silhouette score, Davies-Bouldin, cluster interpretability
Forecasting	MAE, RMSE, MAPE, rolling-window performance

Importance of Model Evaluation

Why Model Evaluation Matters:

- Ensures model reliability and robustness.
- Helps detect overfitting and underfitting.
- Assists in selecting the best model for a given task.
- Provides insights into model performance on unseen data.

Loss Functions and Risk

To "learn," we must quantify error.

单次预测的惩罚。

- **Loss Function** $L(y, \hat{y})$: The penalty for a single prediction.

- **Empirical Risk**: The average loss over the training set:

经验风险：训练集上的平均损失

$$R_{emp}(f) = \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i))$$

- **Principle of ERM**: We choose a function f that minimizes R_{emp} .

经验风险模型 (ERM) 原理：我们选择一个函数 f 来最小化 R_{emp}

Bias-Variance Tradeoff

偏差-方差权衡

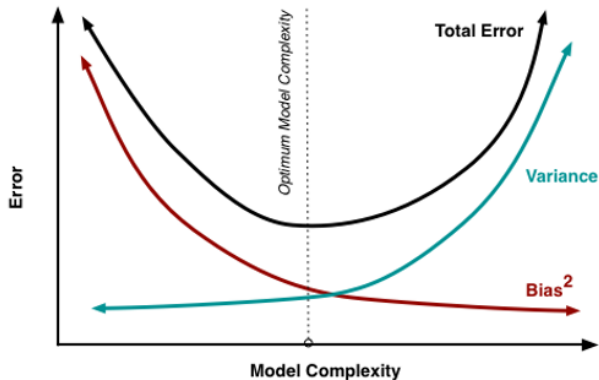
Bias: Error due to overly simplistic assumptions (Underfitting).

偏差：由于过于简单的假设而导致的误差（欠拟合）

Variance: Error due to excessive sensitivity to training data (Overfitting).

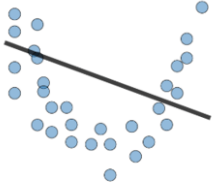
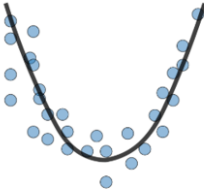
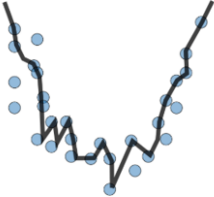
方差：由于对训练数据过度敏感而导致的误差（过拟合）

- **High bias leads to underfitting.** 高偏差会导致欠拟合
- **High variance leads to overfitting.** 高方差会导致过拟合
- The goal is to find an **optimal balance between bias and variance.**
目标是找到偏差和方差之间的最佳平衡点



Source: Dataquest

Underfitting vs Overfitting

	Underfitting	Just right	Overfitting
Symptoms	• High training error • Training error close to test error • High bias	• Training error slightly lower than test error	• Very low training error • Training error much lower than test error • High variance
Regression illustration			

Source: Machine Learning tips and tricks cheatsheet

Underfitting vs Overfitting

	Underfitting	Just right	Overfitting
Symptoms	• High training error • Training error close to test error • High bias	• Training error slightly lower than test error	• Very low training error • Training error much lower than test error • High variance
Classification illustration			

Source: Machine Learning tips and tricks cheatsheet

Underfitting vs Overfitting

	Underfitting	Just right	Overfitting
Symptoms	• High training error • Training error close to test error • High bias	• Training error slightly lower than test error	• Very low training error • Training error much lower than test error • High variance
Deep learning illustration			

Source: Machine Learning tips and tricks cheatsheet

Underfitting vs Overfitting

	Underfitting	Just right	Overfitting
Symptoms	• High training error • Training error close to test error • High bias	• Training error slightly lower than test error	• Very low training error • Training error much lower than test error • High variance
Possible remedies	• Complexify model • Add more features • Train longer		• Perform regularization • Get more data

Source: Machine Learning tips and tricks cheatsheet

Characteristics of a Good Model

- **Accuracy:** Produces results that match real-world observations.
准确性：能够产生与真实世界观测结果相符的结果
- **Generalizability:** Works well with unseen data.
泛化能力：能够很好地处理未见过的数据
- **Interpretability:** Can be easily understood and explained.
可解释性：易于理解和解释
- **Efficiency:** Uses computational resources effectively.
效率：能够有效地利用计算资源
- **Scalability:** Performs well with increasing data.
可扩展性：能够很好地处理不断增长的数据。

Common pitfalls and good practice

常见陷阱

Common pitfalls

- vague objective 目标模糊
- leakage 数据泄露
- poor target definition 目标定义不明确
- wrong metric 错误指标
- ignoring imbalance 忽略不平衡
- overfitting 过拟合

Good practice

- justify the task formulation
阐明任务定义
- inspect data carefully
仔细检查数据
- use train/validation/test correctly
正确使用训练/验证/测试集
- compare with baselines
与baseline比较
- report limitations honestly
报告局限
- keep work reproducible
可复现

Warning

Data leakage occurs when information from the test set "leaks" into the training process.

Common Pitfalls:

- Scaling the whole dataset (calculating global mean/std) before splitting.
- Feature selection using the entire dataset.
- Temporal leakage: Using future data to predict the past.

Summary and next lecture

- Analysis comes before modelling.
- A modelling task must be clearly defined.
- EDA is essential for good feature and evaluation decisions.
- Robust workflows require correct splitting, sensible baselines, and appropriate metrics.
- Later lectures will study specific algorithms in more detail.

Next lecture

Unsupervised Learning: clustering, similarity, and structure discovery without labels.